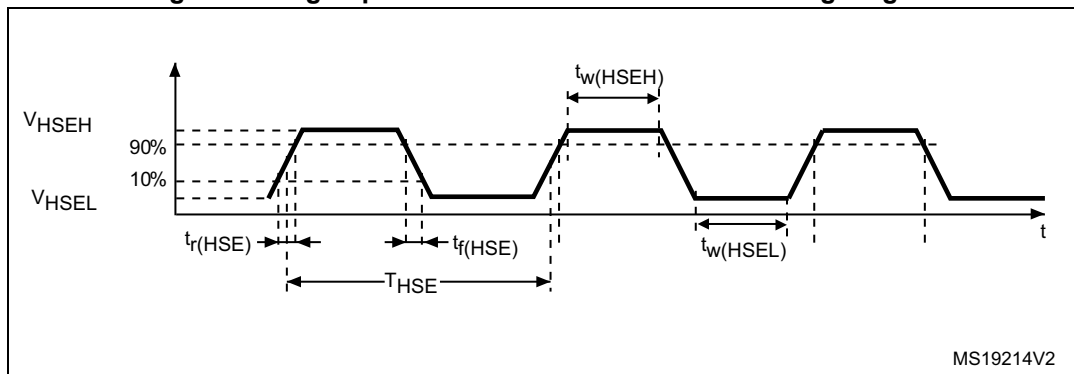


1. Specified by design. Not tested in production.

**Figure 17. High-speed external clock source AC timing diagram**



### Low-speed external user clock generated from an external source

In bypass mode the LSE oscillator is switched off and the input pin is a standard GPIO.

The external clock signal has to respect the I/O characteristics in [Section 5.3.13](#). See [Figure 18](#) for recommended clock input waveform.

**Table 38. Low-speed external user clock characteristics**

| Symbol                    | Parameter <sup>(1)</sup>              | Conditions | Min                    | Typ    | Max                    | Unit |
|---------------------------|---------------------------------------|------------|------------------------|--------|------------------------|------|
| $f_{LSE\_ext}$            | User external clock source frequency  | -          | -                      | 32.768 | 1000                   | kHz  |
| $V_{LSEH}$                | OSC32_IN input pin high level voltage | -          | $0.7 \times V_{DDIO1}$ | -      | $V_{DDIO1}$            | V    |
| $V_{LSEL}$                | OSC32_IN input pin low level voltage  | -          | $V_{SS}$               | -      | $0.3 \times V_{DDIO1}$ |      |
| $t_{w(LSEH)}/t_{w(LSEL)}$ | OSC32_IN high or low time             | -          | 250                    | -      | -                      | ns   |

1. Specified by design. Not tested in production.

**Figure 18. Low-speed external clock source AC timing diagram**

